



DT-BEMS: Digital twin-enabled building energy management system for information fusion and energy efficiency

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ABSTRACT

Building Energy Management Systems (BEMS) collect and monitor building operational data for energy efficiency but face challenges such as complex data management, reliability issues, and a lack of holistic approaches to energy efficiency. To address these limitations, this study proposes Digital Twin-enabled BEMS (DT-BEMS), integrating Digital Twin technology with BEMS to enhance functionality. DT-BEMS uses the Brick schema to create a BEMS ontology and incorporates a virtual building model defined as a mathematical representation of building physical behavior. This model includes three behavioral models: the energy imputation model, the Predicted Mean Vote (PMV) virtual model, and the occupancy model. Holistic Operational Signature (HOS) analyzes building operation data, identifying patterns and relationships for comprehensive efficiency evaluation. In DT-BEMS, a rule-based HOS classifier enables real-time operational signature identification, generating energy efficiency insights. The process of energy efficiency information generation was continuously updated throughout the building operation stages to ensure reliability and accuracy. The proposed system was validated using one month's data from an office space, with data expanded at three-day intervals. DT-BEMS identified eight operational signature types, achieving a 97.9 % accuracy rate when using just 12 days of data compared to a full month. During the study, 23.5 % of operations were identified as inefficient, with energy waste during unoccupied periods being the most prevalent (19.9 %). The virtual model improved identification accuracy by 4.6 % of the data. Real-time operational signature identification was implemented in the Dynamo environment, highlighting DT-BEMS as a practical tool for enhancing building energy management, reducing inefficiencies, and supporting sustainable building operations.

Nomenclatures

Abbreviations

Adam	adaptive moment estimation
AEC	Architecture, engineering and construction
BEMS	build energy management systems
BIM	building information modeling
BMS	building management system
Clo	clothing
DH	district heating
DR	demand response
DT	digital twin
EHP	electric heat pump
GMM	gaussian mixture model
HOS	holistic operational signature
HVAC	heating, ventilating, and air conditioning

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Abbreviations

IoT	internet of things
ISO	international organization for standardization
MAE	mean absolute error
MCMC	Markov chain Monte Carlo
MET	metabolic equivalent task
MLP	multi-layer perceptron
MPC	model predictive control
MRT	mean radiant temperature
O&M	building operation and maintenance
OS	operational signature
PIR	passive infrared
PLM	product lifecycle management
PMV	predicted mean vote
PV	photovoltaic

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Abbreviations	
RDF	resource description framework
ReLU	rectified linear unit
Symbols	
C_f	correction function
C_{res}	sensible heat exchange through respiration
E	evaporation
H	heat loss through convection and radiation
M	metabolic rate
n	number of datasets
x	true value
$\hat{x}$	predicted value
W	effective mechanical work
Subscripts	
c	cooling
i	counter
res	respiration

1. Introduction

In the face of the growing importance of addressing the climate crisis and achieving carbon neutrality, building energy management has emerged as a crucial task for a sustainable future. Approximately 40 % of global energy consumption occurs in buildings, accounting for a significant portion of total carbon emissions [1]. Urbanization and economic growth have led to a continuous increase in building energy consumption, making efficient energy management a necessity rather than a choice [2]. Therefore, improving the energy efficiency of buildings is essential for reducing carbon emissions and mitigating climate change [3]. By effectively implementing energy management strategies, it is possible to lower energy consumption, reduce greenhouse gas emissions, and ultimately achieve sustainable development objectives.

Building Energy Management Systems (BEMS) are important tools used to monitor and control energy consumption within buildings, thereby enhancing energy efficiency [4]. BEMS collect real-time data through sensors and metering devices, analyze energy usage patterns, and propose optimized energy management strategies [5]. These systems integrate various building systems such as heating, cooling, lighting, and ventilation to maximize energy-saving effects. As a result, they reduce energy consumption, lower operational costs, and improve occupant comfort, making BEMS an essential component of modern building management. Furthermore, the information on building energy behavior and efficiency analyzed through BEMS during the building operation phase can significantly contribute to policy and certification systems, urban management, and the advancement of zero-energy building design and engineering in the construction industry [6]. This comprehensive data supports the development of informed energy policies, helps cities manage their energy resources more efficiently, and drives the construction industry towards achieving higher standards of sustainability [7].

However, traditional BEMS have several limitations. Primarily relying on sensor and metering-based monitoring incurs high installation and maintenance costs. The complexity of managing large volumes of data from various sensors can be challenging, and any issues with data accuracy and reliability can compromise the effectiveness of energy management. Additionally, these systems are often limited to data-based analysis, restricting the application of advanced energy efficiency solutions [8]. There is also an insufficient understanding of the interaction between building physical behavior and energy consumption, which limits the potential for maximizing energy management effectiveness. To address these issues, a more sophisticated and integrated approach is necessary. By leveraging Digital Twin technology, which has been actively studied in recent years, the capabilities of BEMS can be enhanced [9]. This approach addresses the limitations of traditional BEMS, enabling more efficient building management.

The concept of the Digital Twin was first introduced at University of

Michigan presentation in 2002 aimed at establishing a Product Lifecycle Management (PLM) center [10]. Digital Twin, initially proposed in the manufacturing industry, has continuously developed and is utilized across various industries for purposes such as monitoring, simulation, optimization, and diagnosis, among others [11]. Recently, a plenty of papers on Digital Twin have been published in the Architecture, Engineering and Construction (AEC) industry [12]. Similar to its use in product lifecycle management in manufacturing, Digital Twin enables effective building management throughout the building lifecycle in construction. In particular, the focus of Digital Twin is on O&M (Operation and Maintenance) stage, where Digital Twin significantly contributes to the operation and management of physical buildings [13].

Digital Twin technology creates a virtual model of physical systems, enabling real-time monitoring and prediction based on real-time data [10]. Tao et al. [11] proposed a five-dimensional Digital Twin concept, defined by physical entities, virtual models, services, and the connections between them. These connections, or relations, encompass the exchange of data, information, and knowledge. By applying Digital Twin technology to BEMS, interaction between the physical building and the virtual model, based on real-time data, becomes possible. From this perspective, integrating Digital Twin with BEMS provides energy efficiency information or knowledge that reflects the building's physical behavior using real-time data. Therefore, combining Digital Twin technology with BEMS can help address certain limitations, such as an overreliance on data-based analysis and a lack of comprehensive understanding of the building's physical behavior and energy consumption.

Some studies have demonstrated the potential for integrating Digital Twin technology with BEMS. Faham et al. [14] explored the application potential of Digital Twin technology in building energy management, highlighting its ability to monitor, optimize, and predict energy consumption in real time, thus contributing to sustainable building energy management and cost reduction. Moreover, Grieves and Vickers [10] have extensively discussed the potential of Digital Twins in enhancing the predictability and optimization of complex systems, which is applicable to building energy management as well. Zakharchenko and Stepanets [15] proposed the concept of Digital Twin as an integrated platform for Building Management Systems (BMS), Building Information Modeling (BIM), and Internet of Things (IoT) in building automation to enhance energy efficiency. Hwang et al. [16] proposed a Digital Twin-based BEMS (DT-BEMS), discovering a potential energy savings of 17.2 % based on data from the past month. These studies underscore the potential of Digital Twin technology to enhance the accuracy and effectiveness of energy management systems. Applying Digital Twin technology to building energy management allows for more accurate energy management through real-time interaction between physical buildings and virtual models. This enables the implementation of advanced applications and significantly improves energy efficiency. Digital Twins can detect system changes and anomalies in real-time, allowing for immediate response, and contributing to long-term energy savings and cost reductions.

Despite the potential of integrating Digital Twin technology with BEMS, the technical development for embedding DT into BEMS remains insufficient, necessitating further studies [17]. The limitations of previous studies are as follows:

- **Absence of a detailed methodology for virtual modeling:** Existing research lacks robust virtual modeling methodologies and improvement processes for virtual models during the O&M stage. Virtual models must be developed in-situ and continuously to accommodate the variability inherent in building operations [18].
- **Lack of data quality management within BEMS:** There is a notable deficiency in strategies to address data missing due to communication errors, sensor failures, and data mapping issues within BEMS. The quality of data acquired from BEMS significantly impacts

applications aimed at improving building energy efficiency, thereby necessitating improvements in data quality.

- **Insufficiency of comprehensive energy efficiency strategies and real-time implementation methodologies:** There is a lack of comprehensive energy efficiency strategies and methodologies for real-time implementation in buildings. Instead of merely reducing absolute energy consumption, it is necessary to adopt a holistic approach that considers the indoor environment from an occupant-centric perspective and incorporates the building's own power generation in a carbon-neutral context. Additionally, methodologies that can apply these energy efficiency strategies in real time to actual buildings are essential.
- **Provision of operational solutions limited to a data-driven approach:** Existing systems propose energy efficiency strategies that are restricted to a data-driven approach. However, meaningful insights into building operational states should be provided by leveraging multi-source data.

The purpose of this study is to propose a Digital Twin-based Building Energy Management System (DT-BEMS) and demonstrate the synergy between Digital Twin technology and BEMS by presenting the concept and implementation methodologies for DT-BEMS. DT-BEMS is a building management system that integrates Digital Twin with traditional BEMS to achieve real-time energy efficiency. The specific objectives of this study are outlined as follows:

- **In-situ and continuous construction of the virtual model:** The in-situ and continuous construction of the virtual model facilitates ongoing adaptation and refinement, effectively reflecting real-time changes in the building's operational conditions. Yoon [19] defined the virtual model in a Digital Twin as one that integrates both geometrical information and a mathematical model to describe the physical behavior of a building throughout its lifecycle. In-situ modeling involves creating more accurate models tailored to the specific operational conditions of the target building, compared to those developed using external factors such as different buildings or modeling environments [20]. This study utilizes operational data from BEMS and intrusive measurement data to construct the virtual model that describes the physical behavior of the building. Additionally, the performance of the virtual model continuously improves as data is accumulated during the building's operational phase.
- **Improvement of BEMS data quality:** This research enhances operational data quality by representing the structure and architecture of BEMS in ontology from an informatics perspective. Moreover, it employs machine learning-based virtual models that reflect the physical behavior of the building.
- **Provision of comprehensive energy efficiency strategies based on holistic operational signature (HOS):** HOS assesses the operational state comprehensively by identifying the operational signature of the building. This study proposes a rule-based HOS classifier to provide holistic energy efficiency strategies in real-time, considering the indoor environment, power generation by the building's facilities, and energy consumption.
- **DT-BEMS from the perspective of building operations informatics:** This study analyzes the patterns of individual multi-source data and applies virtual models to generate meaningful information. These insights are implemented as information on building operational states through HOS-based operational signature identification, providing a foundation for optimal building control. Additionally, DT-BEMS is updated based on the operational data accumulated during its operation, generating more reliable and accurate information and knowledge.

DT-BEMS optimizes energy consumption, reduces maintenance costs, and enhances occupant comfort by processing real-time data and employing advanced applications. This contributes to sustainable

building operations and supports the achievement of carbon neutrality goals. Additionally, this study provides a foundational reference for developing diverse energy management solutions utilizing Digital Twin technology in the future.

The remainder of this paper is composed as follows. Section 2 introduces the methodology for identifying operational signatures in DT-BEMS, while Section 3 explains how this method was applied to real BEMS. Section 4 includes analytics of the results.

2. Digital Twin-based building energy management system

2.1. DT-BEMS concept

DT-BEMS is an advanced energy management system from the perspective of operational informatics that leverages real-time data from BEMS to create a virtual model that reflects the dynamic characteristics of a building throughout its lifecycle and provides holistic energy efficiency strategies in real-time. The concept of DT-BEMS proposed in this study is shown in Fig. 1. DT-BEMS enables the development of energy efficiency strategies through (1) BEMS informatics based on ontology construction, (2) application of virtual models to generate behavior-reflective data, (3) real-time operational signatures identification by HOS, and (4) continuous updates on DT-BEMS across building operations. BEMS informatics is designing a metadata schema of BEMS by constructing BEMS ontology on Brick schema [21]. Virtual models that describe the energy and physical behavior of buildings are generated by leveraging operational data from BEMS. In the case of specific virtual models, the performance can be improved by implementing intrusive measurements, which require the temporary installation of sensors within the building. After constructing the virtual model, the HOS technique is employed as a comprehensive analytical method to provide energy efficiency information. HOS analyses the operation of the building system during the operational stage by identifying operational signatures [22]. This study expands the existing HOS to real-time, enabling real-time identification of operational signatures that comprehensively incorporate multi-source data. Updates on DT-BEMS are performed to reflect the building data more accurately. This approach enables the development of energy efficiency strategies considering the variability in building operations.

2.2. Methods

2.2.1. BEMS informatics

BEMS informatics is defined as the representation of a building's architecture and the BEMS data collection system in the form of ontology from an informatics perspective. BEMS informatics enables a more precise understanding of the BEMS system and provides a foundation for efficient building system management. Additionally, it addresses the mapping issues within BEMS data by representing the data collection system, thereby improving the quality of the collected data. BEMS ontology represents the control points and data collection processes within BEMS by leveraging Brick schema. Brick schema is a metadata schema that standardizes semantic descriptions of the physical, logical, and virtual assets in buildings and the relationships between them [23]. Brick schema comprises three main entities:

- **Location:** represents physical spaces such as floor, space, or zone
- **Equipment:** covers various devices like a meter, AHU, and chiller
- **Point:** includes specific points that generate or utilize data, including sensors, setpoints, and status indicators

By using these entities to represent BEMS control points and defining the relationships between them, BEMS ontology offers a clear and structured view of the BEMS. This structured representation enhances system understanding and supports effective building energy management.

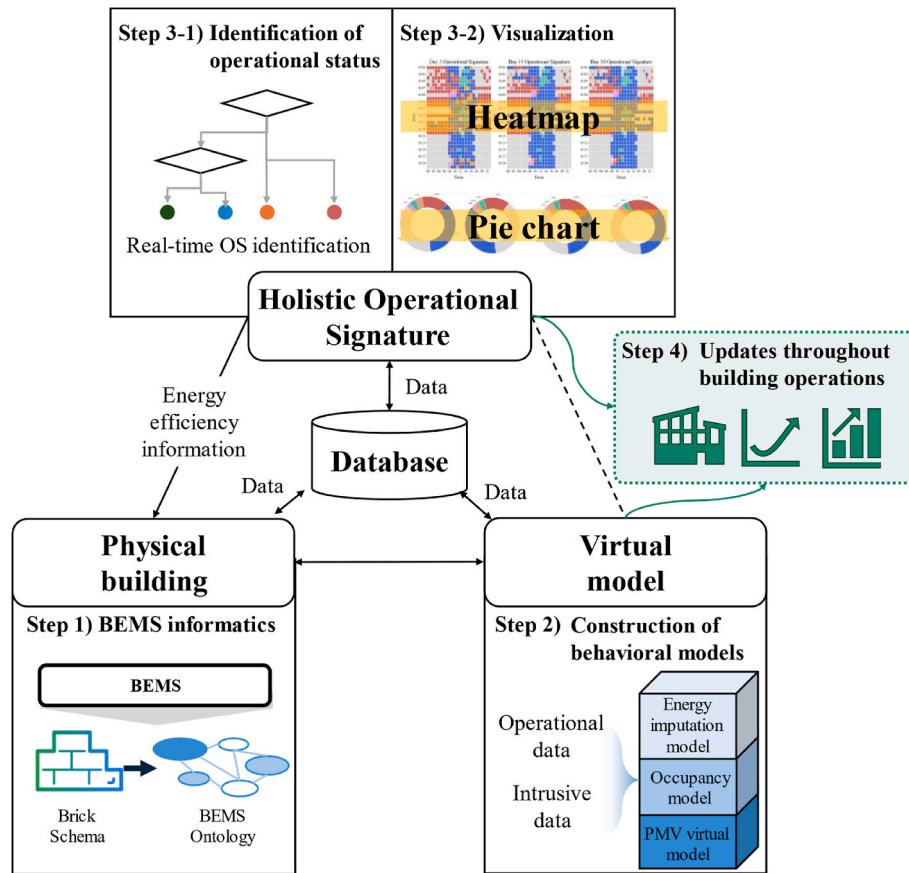


Fig. 1. Digital Twin-enabled BEMS concept.

2.2.2. Virtual model

Virtual model is a mathematical model that represents the physical behavior of building throughout its lifecycle [19]. By applying virtual model, it is possible to predict variables that are challenging to measure with physical sensors, enhance the quality of data collected within buildings, and generate data that reflects the physical behavior of buildings. In DT-BEMS environments, the virtual model includes three types of behavioral models: energy imputation model, occupancy model, and predicted mean vote (PMV) virtual model.

The energy imputation model is defined as a data imputation method for building energy variables, developed to predict energy consumption and generation throughout the building lifecycle. One of the major issues related to the quality of BEMS data is the frequent occurrence of missing values. These missing values degrade the quality of data collected by BEMS, significantly impacting building energy analysis. To address this issue, DT-BEMS applies a machine learning-based energy imputation model, filling in missing values with data that reflects the building's physical behavior. This approach directly enhances the quality of BEMS data.

The occupancy model is an occupant detection model used to infer the presence of occupants in specific spaces when occupant-detecting physical sensors are unavailable. Building occupancy is crucial for the efficient control of building systems and for reducing energy consumption based on occupancy status [24]. While machine learning can be a method for occupancy modeling in environments where physical sensor-based occupant detection is limited, obtaining ground truth of occupancy data remains a significant challenge [25]. Furthermore, buildings with clear purposes, such as office or residential buildings,

generally follow typical operational schedules. Consequently, this study leverages operational schedules and operational data analysis to construct the occupancy model.

The PMV virtual model is a gray-box model designed to predict the PMV for real occupied area. Among various indoor environmental factors, the thermal environment is the one most directly related to building system energy usage [26,27]. PMV is an index established by Fanger for the evaluation of thermal environments [28]. PMV is a widely used thermal environment assessment index, and its calculation is formula-based, making it suitable for virtual model development and facilitating measurement using physical sensors [29]. For these reasons, DT-BEMS utilizes the PMV index for indoor environment evaluation. It ranges from -3 to $+3$, where values approaching -3 indicate colder conditions, values nearing $+3$ denote warmer conditions, and values approximating 0 signify a state of comfort. Additionally, since indoor environmental sensors are usually installed on walls, it is difficult to accurately assess the thermal environment of the central areas where people reside based solely on the variables measured by these sensors. Therefore, developing the PMV model is essential for understanding the thermal environment of real occupied areas and conducting comprehensive energy efficiency analysis. Kim and Yoon [30] implemented a model to predict the PMV of the real occupied zones using intrusive data depending on the modeling method. Furthermore, they highlighted that a gray box virtual model, which corrects PMV values calculated using the ISO 7730 standard [31] through Bayesian estimation, showed superior predictive performance. The gray-box PMV virtual model necessitates intrusive measurements to gather PMV data of the target area, requiring temporary installation of PMV sensors.

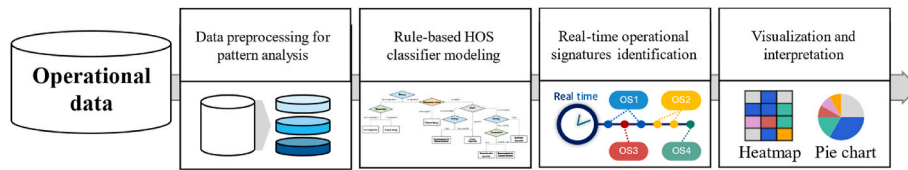


Fig. 2. Process of HOS application in DT-BEMS.

2.2.3. Holistic operational signature

The real-time provision of energy efficiency information in DT-BEMS is achieved based on the HOS by utilizing the results of the virtual model and the operational data from BEMS. HOS is a technique for identifying the relationships and patterns among various variables during the building operation stage, based on operational signature type analysis. It enables a comprehensive efficiency analysis of building systems. Operational signature is an analytical approach that reveals unique correlations between building energy variables and environmental variables. Operational signature is useful for identifying energy consumption patterns, operational conditions, and potential abnormal behaviors. Hong et al. [22]. Introduced the HOS concept, including multiple operational signature elements (x-OS) and representation types, for comprehensive operational analysis of building-side district heating (DH) substations. By applying HOS, it is possible to conduct post-evaluations of building design during the operational phase, analyze control quality within building systems, and assess energy-saving potential. In DT-BEMS, the process of real-time operational signature identification and energy efficiency information provision through HOS is illustrated in Fig. 2. This includes data preprocessing for pattern analysis through symbolic representation and modeling a rule-based HOS classifier which performs real-time identification, visualization, and interpretation of operational signatures. The process of data preprocessing encompasses the assignment of meaning to numerical data and is essential prior to HOS-based data analysis. To enable clustering-based HOS for real-time applications, a rule-based HOS classifier is required to identify operational signatures at each data collection interval. HOS classifier utilizes data such as building energy usage, solar power generation, and indoor environmental PMV to identify the current state according to predefined operational signature types. Upon completing the identification, energy efficiency insights are generated through visualization and analysis. For building energy usage, heating and cooling energy consumption are primarily analyzed as they have the most significant impact on indoor environmental conditions and represent the largest portion of energy consumption in buildings. The HOS classifier is subsequently applied to provide a comprehensive analysis of building energy efficiency, supporting informed decision-making for energy management and optimization.

2.2.4. Updates on DT-BEMS

DT-BEMS is continuously updated as data accumulates throughout the building operation phase, enabling the provision of energy efficiency strategies that more accurately reflect building operational data. This approach is designed to ensure that DT-BEMS can adapt to the variability in building operations to generate more accurate and reliable information and knowledge from an informatics perspective. This

process involves continuously incorporating building operational data into the virtual model and HOS. The virtual model should be continuously updated as operational data from BEMS accumulates to reflect the characteristics of physical building from an in-situ, site-centered modeling perspective. For example, if an occupancy model uses indoor CO₂ concentrations to determine occupancy, the results vary as the CO₂ threshold defining occupancy is consistently updated. Modifications to the HOS occur during the data preprocessing stage as the criteria for symbolizing data are continually adjusted throughout the building's operational phases. These changes subsequently refine the outcomes of the HOS.

3. Application of Digital Twin-based BEMS

3.1. Target space

A three-story building is utilized for the application of the DT-BEMS as shown in Fig. 3. DT-BEMS environment was built for second floor office space. A detailed description of the target building is provided in Table 1. The target building employs a heating and cooling system based on the Electric Heat Pump (EHP). The building's energy generation system employs PV (photovoltaic) panels for solar power production. Building data utilized includes one month of BEMS data (from January 2021, recorded at 5-min intervals) and five days of intrusive measurement data. The intrusive measurement data were specifically used to develop the PMV virtual model, thereby deriving the PMV of the actual occupied area. DT-BEMS was implemented within the Dynamo environment, enabling the real-time identification of operational signatures and determination of the operational status. Therefore, holistic energy efficiency information of building systems was created. Furthermore, the operational signatures identification was progressively updated to achieve more accuracy as data accumulated every three days.

Table 1
Description of the target building.

Element	Description
Location	Gyeongbuk, South Korea
Building usage	Public service facility
Target space	Office area on 2F
Heating, ventilating, and air conditioning (HVAC) system	- 2 packaged air conditioner units (EHP) - A mechanical ventilation unit
Energy generation	PV panel-based solar power generation system

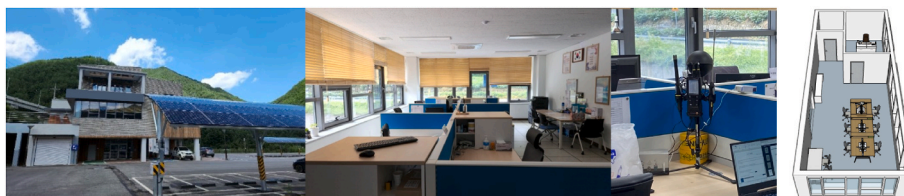


Fig. 3. Target space and intrusive data acquisition images.

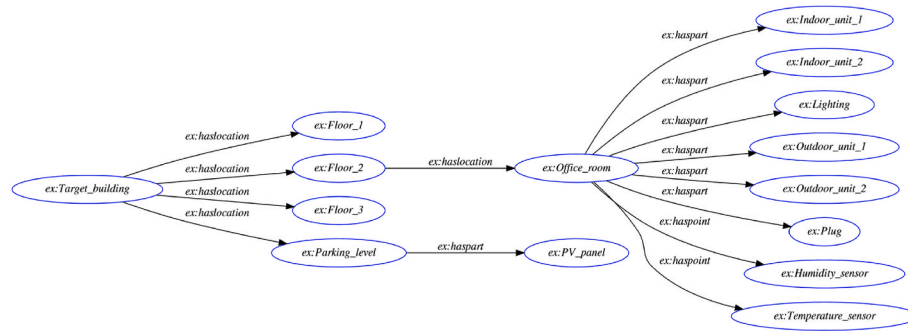


Fig. 4. BEMS ontology of target space.

3.2. Target BEMS informatics

BEMS ontology of the target building was represented using the Brick schema, with a focus on the second-floor office space. The building consists of three floors and a parking lot, where solar power is generated by PV panels. Energy consumption in the second-floor office space is divided into three main components: heating and cooling, lighting, and plug. This study primarily focused on heating and cooling usage, which relies on two indoor units and two outdoor units. An indoor environmental sensor in the target space collects temperature, humidity, and CO₂ data.

As shown in Fig. 4, the architecture of the BEMS system was designed using the Resource Description Framework (RDF) and then converted into a TTL file for visualization. The development of the BEMS ontology facilitated the integration and interpretation of data from multiple sources, establishing a foundation for comprehensive energy efficiency strategy development for building managers. Additionally, the BEMS ontology addresses data mapping issues by clearly defining control points within the BEMS.

3.3. Application of the virtual model

3.3.1. Energy data imputation model

To fill missing values in the energy dataset of the target BEMS, an energy data imputation model based on an MLP (Multi-Layer Perceptron) [32] was developed. For the HOS application, models were developed for building heating and cooling energy consumption data as well as solar power generation data. In the target BEMS system, there are four types of cooling and heating energy data: the usage of outdoor units 1 and 2, and the usage of indoor units 1 and 2. However, the usage of indoor units is minimal and infrequently measured. As a result, energy data imputation models were developed for outdoor units 1 and 2, along with solar power generation. The model structure and input variables for each model are presented in Table 2.

To address missing data for the next three days, the three types of models were re-trained every three days as new data accumulated. For

example, data from days 1–3 was used to impute missing values for days 4–6. The imputed data from days 4–6 was then combined with the data from days 1–3 to create a dataset covering days 1–6. The model trained on data from days 1–6 was subsequently used to address missing data for days 7–9.

3.3.2. Occupancy model

The process of occupancy model consists of an initial occupancy state setting based on the operational schedule and occupancy updating by using operational data. In operational data, building energy consumption and indoor environmental variables can be utilized in the occupancy model. However, if energy consumption is used for occupancy estimation, it becomes difficult to detect energy waste in unoccupied situations. Additionally, among the environmental variables measured by indoor sensors, CO₂ concentration is most strongly correlated with occupancy status [33,34]. Therefore, as illustrated in Fig. 5, this study constructed an occupancy model based on the operational schedule and CO₂ concentration.

Since the target space is an office environment, daytime was set as the initial occupied state, and nighttime was set as the unoccupied state. In the initial occupied state, if the absolute CO₂ concentration is low (below the average value during occupied periods) and shows a decreasing trend, it was updated to unoccupied state. In contrast, if the CO₂ concentration during initial unoccupied state is higher than the average CO₂ concentration during unoccupied time and increasing, the status was changed to occupied. As the average CO₂ concentration changes with data accumulation, the occupancy model result was updated every three days.

3.3.3. PMV virtual model

For the PMV virtual model construction, three days of intrusive data were used. The modeling approach employed the Bayesian MCMC (Markov Chain Monte Carlo) [35] based gray box modeling method proposed in a previous study [30]. In this modeling approach, a white box modeling process, utilizing the PMV equation (Equation (1)) to calculate PMV, precedes the correction process through Bayesian

Table 2
Settings of energy imputation model.

Model type	Energy consumption	Solar power generation
Input variables	energy consumption from 5 min prior, outdoor temperature, indoor temperature, humidity, carbon dioxide concentration	solar power generation from 5 min prior, outdoor temperature, indoor temperature, humidity, carbon dioxide concentration
Output variable	Energy consumption (outdoor unit)	Solar power generation
Hidden layers (Number of nodes for each layer)	2 layers (64–64)	
Activation function	ReLU	
Batch size/Epoch	32/200	
Loss function	Mean squared error	
Optimizer	Adam	
Ratio of training data set to validation data set	0.8/0.2	

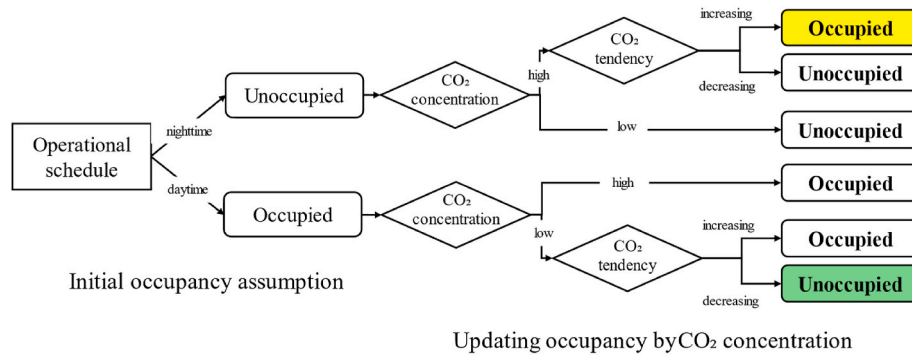


Fig. 5. Occupancy determination process by occupancy model.

MCMC.

$$PMV = (0.303e^{-0.0036M} + 0.28) \times [(M - W) - H - E_c - C_{res} - E_{res}] \quad (1)$$

(1) Setting Variables

To calculate PMV, values of variables such as C_{lo} , Met , MRT , air velocity, dry-bulb temperature, and humidity are required. Among these, dry-bulb temperature and humidity values were obtained through the data from BEMS. Other variables were assumed based on the criteria outlined in the ISO 7730 standard [31] as shown in Table 3. Considering the characteristics of an office, MET value of 1.0 was used, representing a person sitting comfortably. For C_{lo} , a value of 1.0 was utilized to reflect seasonal variations in the data. Since MRT cannot be directly measured, it was assumed to be the same as the dry-bulb temperature (data from BEMS). Air velocity was set to 0.1 m/s based on common assumptions. Any errors resulting from these assumptions can be minimized through the correction process using Bayesian MCMC after the calculations with the PMV equation.

(2) Correction function

To determine the correction function, constant, 1st-order, 2nd-order, and 3rd-order equations were compared to assess the accuracy. For evaluating accuracy, MAE (Mean Absolute Error) as shown in equation (7) was utilized. As indicated in Table 4, calibration with 2nd-order equation yielded the lowest MAE value. Additionally, as illustrated in Fig. 6, the quadratic correction function provided the most accurate adjustment with the true values. Based on these results, the 2nd-order equation (C_{f2}) was adopted as the form of the correction function. Through Bayesian MCMC, results as Table 5 for the correction parameters (a , b , c) were obtained. Among these, the median values were used to set the coefficients of the correction function.

$$C_{f0} = a \quad (3)$$

Table 3
PMV calculation variables assumptions.

Variables	Value	Meaning
MET	1.0	Seated, relaxed
C_{lo}	1.0	Typical indoor winter clothing ensemble
MRT	Dry-bulb temperature	–
Air velocity	0.1	–

Table 4
Comparison of MAE by order of correction function.

Correction function	MAE after calibration	Increase in MAE before and after calibration
C_{f0}	0.0446	0.1984
C_{f1}	0.0410	0.2019
C_{f2}	0.0384	0.2046
C_{f3}	0.0666	0.1763

$$C_{f1} = aX + b \quad (4)$$

$$C_{f2} = aX^2 + bX + c \quad (5)$$

$$C_{f3} = aX^3 + bX^2 + cX + d \quad (6)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |x_i - \hat{x}_i| \quad (7)$$

3.4. HOS application

The application of HOS involves binning data to symbolize each segment, followed by identifying the operational signature type using the HOS classifier.

3.4.1. Data binning

Data binning in DT-BEMS refers to the process of segmenting various data of BEMS to assign meaning to each segment. Data binning is valuable for simplifying data, allowing for a holistic consideration of diverse datasets. In this study, data binning was performed on three types of data. Absolute criteria were applied to PMV data obtained through the application of the PMV virtual model, while relative criteria were used for energy consumption data and solar power generation data.

(1) Absolute criteria

Data binning using absolute criteria is a method that performs segmentation based on fixed boundaries according to predefined standards. PMV data was binned using absolute criteria based on its definition. The PMV range of -0.5 to 0.5 was defined as the comfort zone, and the data was divided into three segments based on this range. If the PMV value was below -0.5 , it was labeled as cold, and if it was above 0.5 , it was labeled as hot. Since the data used in this study was collected during the winter, PMV values below the comfort zone (cold) were interpreted as indicating an uncomfortable state, while PMV values above the comfort zone (hot) were considered as an over-comfort state.

(2) Relative criteria

Data binning using relative criteria refers to the method of establishing data intervals based on the distribution of existing data values. Due to the significant variability of BEMS energy data, it is challenging to establish fixed boundaries. Therefore, using relative criteria is advantageous. For the binning of energy consumption and solar power generation data, GMM (Gaussian Mixture Model) and K-means clustering were utilized as shown in Fig. 7. GMM was used to identify the boundary between non-operating and operating states. For the operating state, K-means clustering was then applied to determine the boundaries for categorizing the data into three levels: low, middle, and high. Based on these boundaries, energy consumption data and solar power generation data were classified into four clusters. To continuously update the boundaries and better reflect the characteristics of the data,

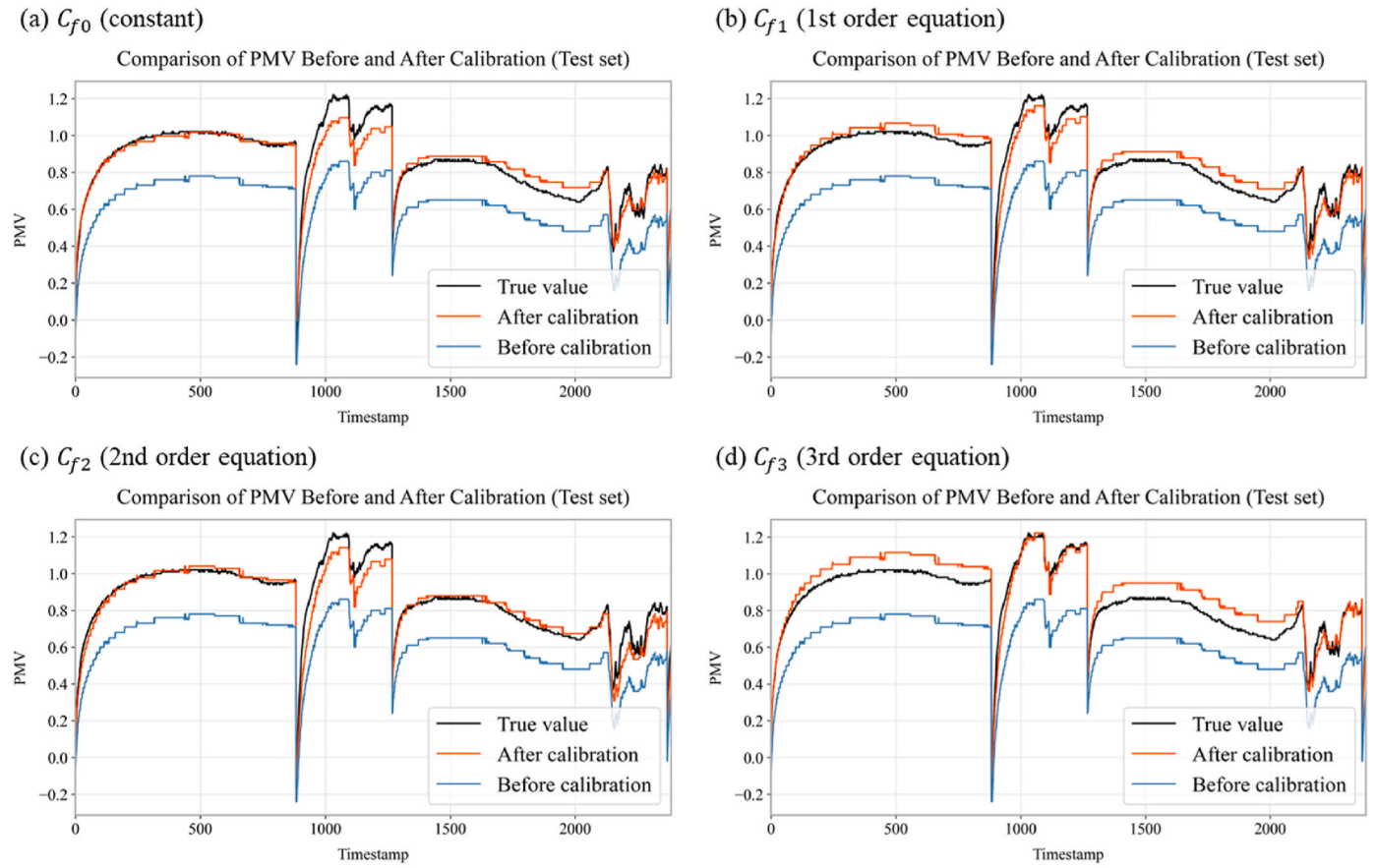


Fig. 6. Comparison of PMV before and after calibration using four correction functions.

Table 5
Correction parameters result.

c			b			a		
Mean	Median	Standard devitaion	Mean	Median	Standard devitaion	Mean	Median	Standard devitaion
0.1534	0.1383	0.3899	0.1039	0.1054	0.4069	0.0274	0.0406	0.3946

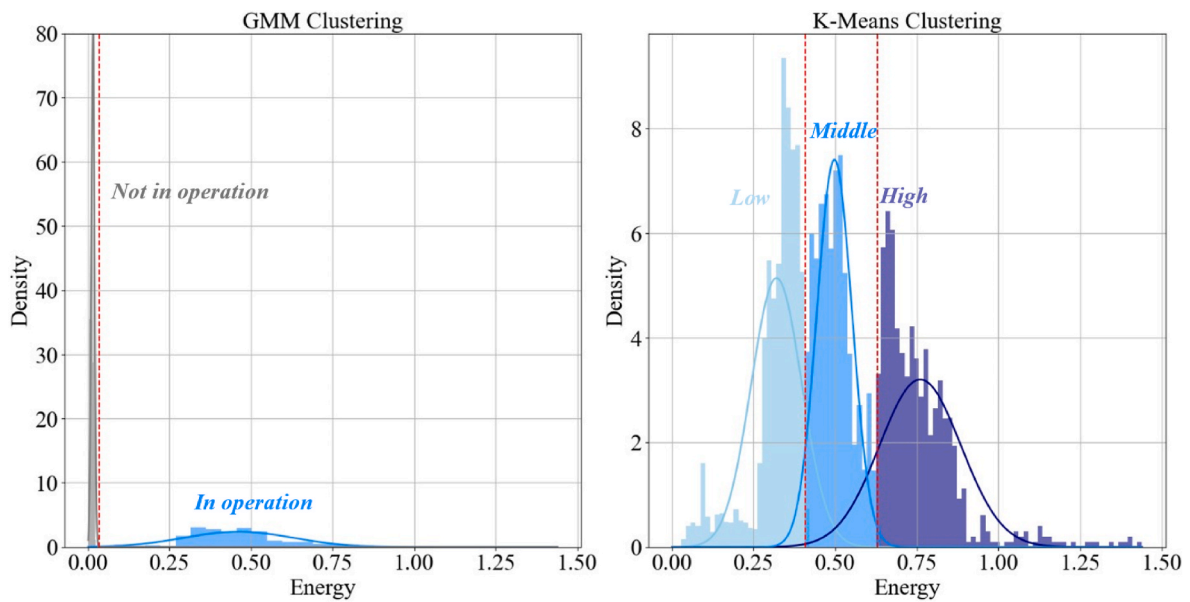


Fig. 7. Boundary values extraction through clustering.

this process was repeated every three days as new data accumulated. For example, the boundaries from the binning process for days 1–3 were used as the criteria for binning days 4–6 data. Additionally, the boundaries from days 1–6 data were used to perform binning for the data from days 7–9.

3.4.2. Operational signatures identification using HOS classifier

After applying data binning, operational signatures for each time were identified in real-time using the rule-based HOS classifier shown in Fig. 8. The types of operational signatures are defined as shown in Table 6. Initially, the operational status of the building system is determined using threshold values of energy consumption obtained from data binning. When the building system is in operation, the occupancy model is applied to ascertain whether the space is occupied. If the target space is unoccupied, it is classified as **Energy waste**. Given that the target area was heated, employing techniques such as radiant heating may imply that energy consumption during periods of unoccupancy is not inherently inefficient. Nevertheless, as the current target area is being heated by the EHP, considerable energy usage during unoccupied periods was deemed wasteful consumption. If the target space was occupied, the operating status is then categorized into one of five categories based on PMV, energy consumption, and solar generation data. For example, a situation where the PMV exceeds 0.5 with high energy usage indicates excessive heating. This condition is classified as **Recommendation for reduced operation**. Conversely, if the PMV is below -0.5 (indicating cold and uncomfortable) and energy usage is high, it suggests that the heating load is excessive, resulting in classification as **Inefficient operation**.

Since the results of behavioral models and data binning were updated every three days, the identification outcomes also followed a three-day update cycle. Datasets accumulated in three-day intervals were used to identify the full month's data, and the results of each dataset were compared. Additionally, a comparison was made between the results obtained before and after applying PMV virtual model and occupancy model.

3.5. Updates on implemented DT-BEMS

Table 7 summarizes the process of updating the virtual model and HOS through data accumulation in three-day intervals, as illustrated in Fig. 9. The virtual model, excluding the PMV virtual model, was updated every three days as data accumulated, aiming for continuous

Table 6
Types of operational signatures.

Current state	Operational signature type	Explanation
The building system is not in operation.	Non-Operation	The building system is in an idle state with very low energy consumption and relatively low solar power generation.
	Surplus Energy	The building system is not in operation, and high solar power generation occurs, resulting in surplus energy.
The target space is unoccupied.	Energy waste	It is an energy waste situation where a significant amount of energy consumption occurs despite the space being unoccupied.
The target space is occupied.	Uncomfortable Operation	This type arises when additional operations are needed to achieve comfort based on PMV but cannot be performed due to insufficient solar power generation, leading to a prolonged state of discomfort. Similar to the Uncomfortable Operation type, this type is characterized by PMV indicating discomfort. However, with relatively high solar power generation, additional operations are proposed to enhance indoor comfort, even if they may result in increased energy consumption.
	Recommendation for Increased Operation	
	Proper Operation	It represents an optimal operational state where PMV is comfort state and energy consumption is at an appropriate level.
	Recommendation for Reduced Operation	This type occurs when the indoor environment exceeds the comfort range, yet the system continues to operate excessively, requiring a reduction in operation.
	Inefficient Operation	Inefficient Operation refers to a state where PMV indicates discomfort despite high energy consumption.

performance enhancement over time. The energy imputation model was updated by remodeling with an incrementally expanding dataset at

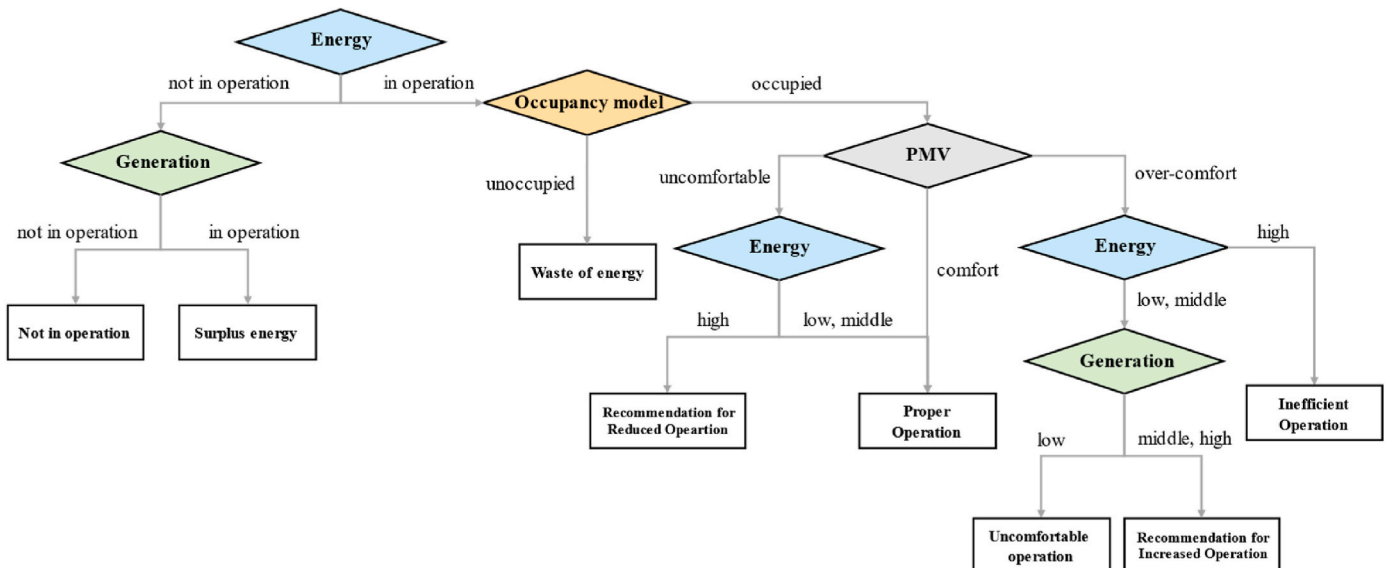


Fig. 8. HOS classifier for operational signatures identification.

Table 7
Types of operational signatures.

Type	Category	Update	Method of update
Virtual model	Energy imputation model	O	Remodeling
	Occupancy model	O	Adjustment of average CO ₂ value
	PMV virtual model	X	–
HOS	Data binning	O	Modification of thresholds for grouping energy consumption and solar generation data
	HOS classifier	X	–

three-day intervals. Regarding the occupancy model, although the initial estimation of occupancy based on the operational schedule remained constant, the subsequent occupancy update outcomes derived from the CO₂ concentration were modified due to changes in the average CO₂ value. The PMV virtual sensor, modeled based on intrusive measurements, was not updated as no additional intrusive measurements were conducted. In the case of HOS, the algorithm of the HOS classifier remained unchanged. However, as the relative criteria-based data binning (clustering-based) for energy consumption and solar generation data was updated every three days, the overall HOS results were also affected.

3.6. DT-BEMS in Dynamo interface

The DT-BEMS environment was developed using the Dynamo interface with a focus on real-time application. Fig. 10 represents the workflow of DT-BEMS for real-time operational signature identification. Energy data and indoor environment data were integrated, and three behavioral models within the virtual building model were applied. After reflecting physical building behavior through the virtual building model, data binning was conducted to symbolize the operational data. Subsequently, the HOS classifier determined the operational signature and energy efficiency information was generated. Operational signature was determined for each timestamp in real-time, and after a one-day analysis, the data was visualized using tools such as heatmaps and pie charts.

4. Results and discussion

4.1. Virtual building model results

4.1.1. The result of energy data imputation model

Three types of models (outdoor unit 1, outdoor unit 2, and solar power generation) were developed at three-day intervals, and their accuracy was evaluated using the R-square metric. Fig. 11 illustrates the accuracy of models that were trained with data accumulated at three-day intervals, and all models showed an increasing trend in accuracy as the data accumulated. For outdoor unit 1, the R-squared value increased from 0.792 to 0.885. Outdoor unit 2 demonstrated the largest improvement, ultimately reaching an R-squared value of 0.973. In the solar power generation model, a high overall accuracy was observed

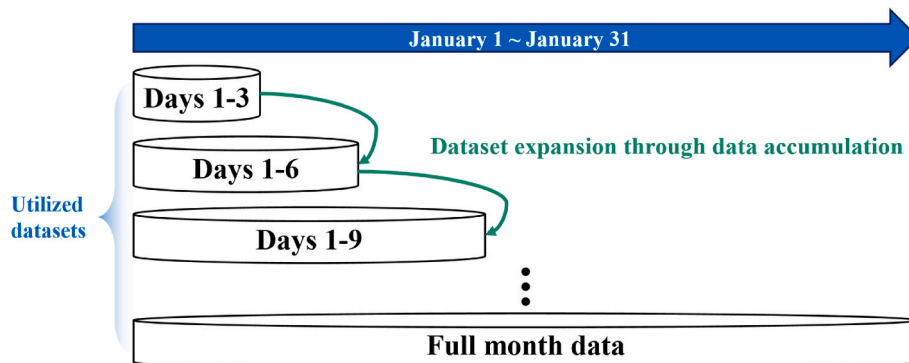


Fig. 9. Dataset utilized based on data accumulated in three-day intervals.

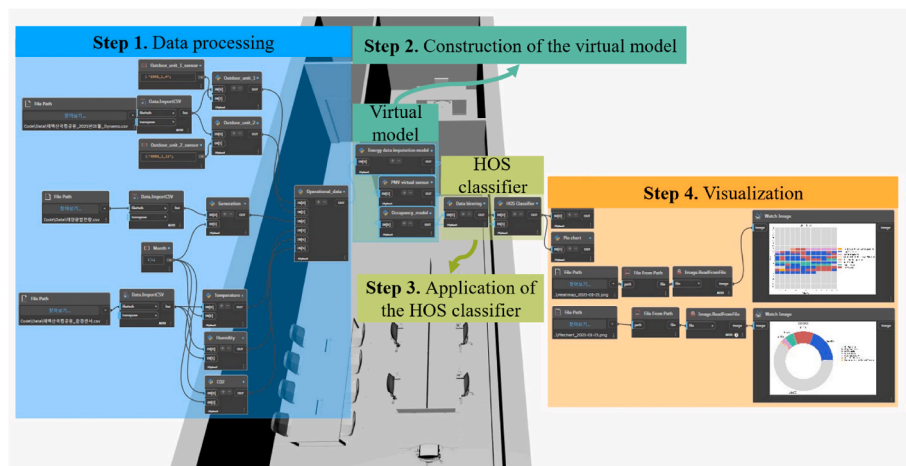


Fig. 10. Implementation of DT-BEMS in the Dynamo environment.

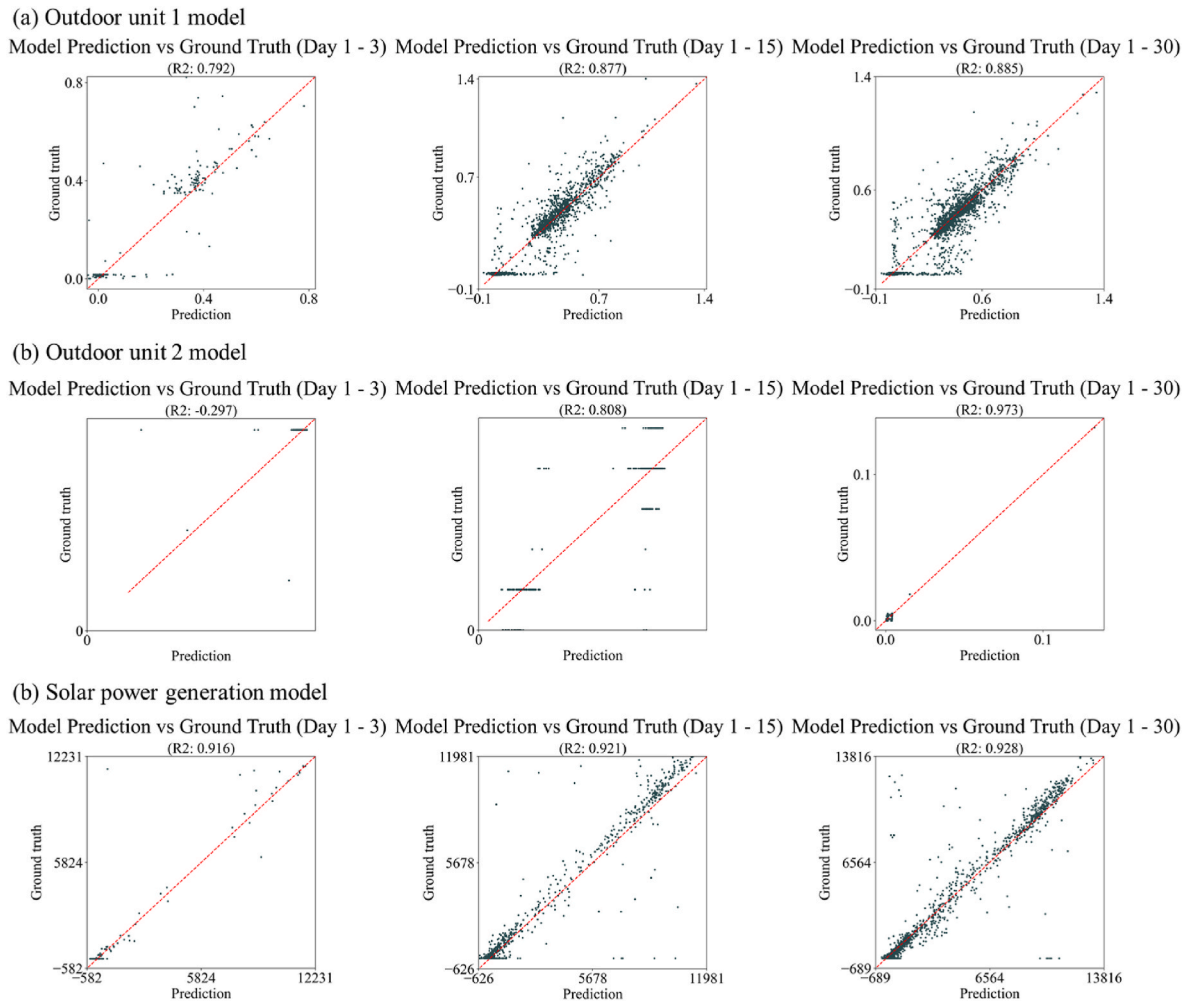


Fig. 11. Energy data imputation model accuracy as data accumulated (test set).

with a gradual increase in accuracy over time. By remodeling the energy imputation model every three days, it gains the advantage of enhanced accuracy as the DT-BEMS operates for longer durations and collects more data.

In the target BEMS system, missing data occurred as follows: 0.12 % for outdoor unit data 1 and 2, and 0.03 % for solar power generation data during the month. Using the energy imputation model, it was possible to fill in missing values with data reflecting the energy behavior of the building in real-time whenever missing data was detected.

4.1.2. Results of the occupancy model

Based on the occupancy model described in Section 3.3.2, the occupancy in the target space was determined. For the month of January, daytime was initially set as occupied and nighttime as unoccupied. Subsequently, occupancy status was updated based on CO₂ values. The daily CO₂ concentration trend for the entire month is illustrated in Fig. 12(a). The line chart in Fig. 12(b) represents the average CO₂ concentrations during occupied and unoccupied times which were initially determined based on the operational schedule. The bar chart in

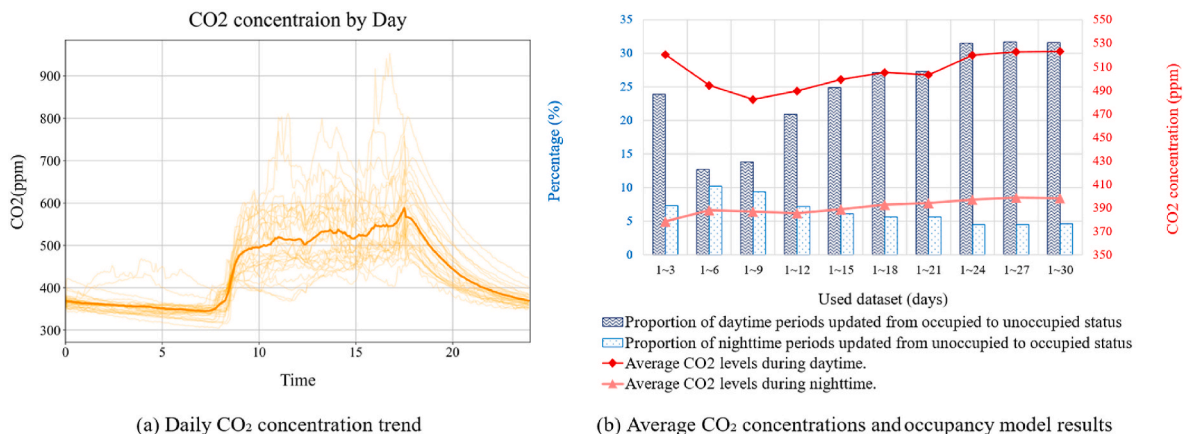


Fig. 12. CO₂ concentration information and occupancy model results.

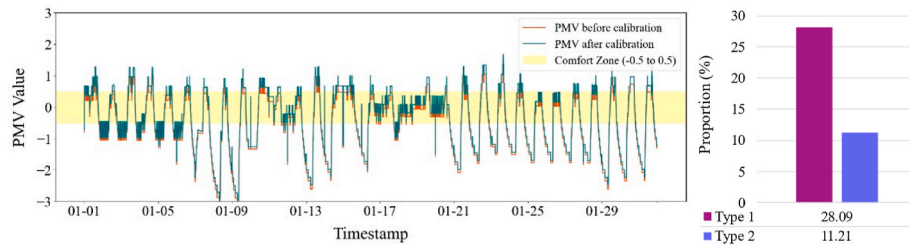


Fig. 13. Difference between before and after PMV model application (Type 1 is initially within the comfort range (-0.5 to 0.5) that shifted outside after applying the PMV model, Type 2 is initially outside the comfort range that moved inside after applying the PMV model.).

Fig. 12(b) illustrates the results of occupancy state updates based on CO_2 concentrations. As the average CO_2 concentration changed, the results of the occupancy state update based on the occupancy model also varied with the accumulation of the dataset. The proportion of initially occupied states updated to unoccupied states by the occupancy model increased with data accumulation, reaching 31.6 %. In contrast, the proportion of initially unoccupied states updated to occupied states showed a decreasing trend as more data was collected.

4.1.3. Results of the PMV virtual model

Using temperature and humidity data from BEMS along with intrusive data, the PMV virtual model was developed to predict the real

occupied area PMV. As shown in Fig. 13, the calculated PMV based on indoor environment sensor data (orange) was calibrated using the PMV virtual model (green). The application of the PMV virtual model resulted in adjustments with a maximum of 0.348 and an average of about 0.129. This indicates a significant difference between the PMV values calculated using the temperature and humidity data from BEMS with the PMV formula (Equation (1)) and the real occupied area PMV obtained through the PMV virtual model application. The bar chart in Fig. 13 represents two scenarios: the proportion of cases that were within the comfort range (-0.5 to 0.5) in the initial calculation but fell outside the comfort range after applying the PMV model (purple), and the proportion of cases that were outside the comfort range in the initial

Data binning results for full month of energy consumption and solar generation data

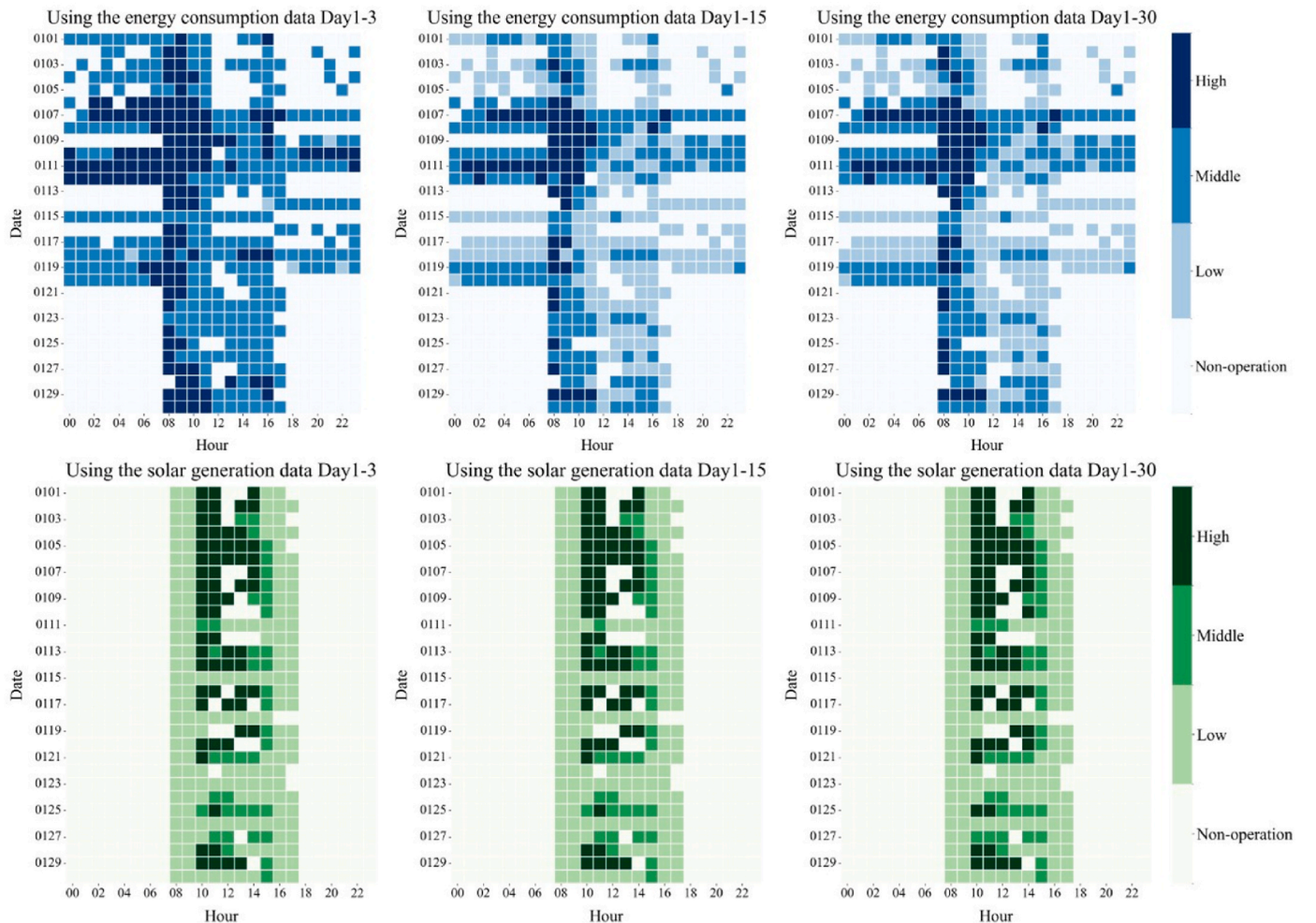


Fig. 14. Data binning using relative criteria.

calculation but entered the comfort range after applying the PMV model (blue). 28.09 % of data fell outside the comfort range that was within the comfort range before applying the virtual model. Conversely, 11.21 % of the data that was originally outside the comfort range moved inside it after applying the virtual model. This indicates that the PMV virtual model not only calibrates PMV values but also enables a more accurate assessment of the actual comfort level in the occupied area. Additionally, PMV values generally increased after applying the PMV model, suggesting that the real occupied areas in the target building tend to have relatively high PMV values due to heat generated by occupants and equipment.

4.2. Operational signatures identification by HOS

4.2.1. Updates based on data accumulation at 3-day intervals

(1) Data binning results

The data binning method described in Section 3.3.1 was applied to energy consumption, solar generation, and PMV data. In contrast to PMV, which was segmented using predefined comfort ranges, energy consumption and solar generation data were categorized with relative criteria based on K-means and GMM clustering. As a result, as datasets were expanded every three days, the boundary values for each segment were updated. Fig. 14 presents the segmentation results of one month of energy consumption and solar generation data when data was accumulated for three days, fifteen days, and a full month. Since the boundary values defining high, middle, low, and non-operation states in the heatmap constantly change, different results were obtained when using the same dataset. Fig. 15 shows a heatmap of PMV data segmented using absolute criteria.

(2) The result of operational signatures identification using HOS classifier

Operational signatures were identified for the entire month's data using datasets accumulated at three-day intervals. The results of these identifications varied with each 3-day data accumulation due to updates in the energy imputation model, occupancy model, and data binning with relative criteria. Fig. 16 depicts the degree of agreement between

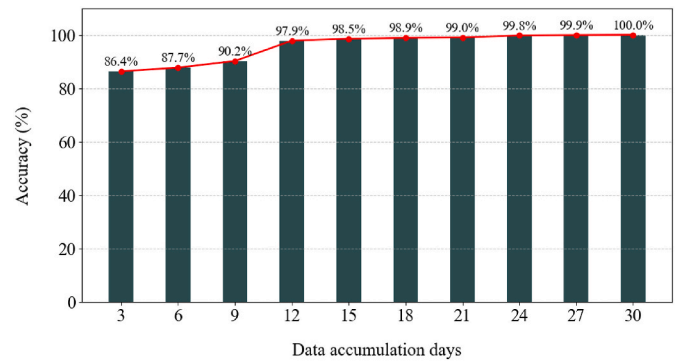


Fig. 16. Comparison of accuracy between verification results using 3-day cumulative data and those using full-month data for verifying the entire month's data.

the operational signatures identified using data accumulated every three days and those identified with the full month's data. As shown in the figure, once 12 days of data are accumulated, the results achieve a 97.9 % agreement with the identification based on the entire month's data. This finding demonstrates that highly accurate results can be obtained even with a reduced dataset, enabling the development of building energy optimization strategies through HOS implementation in a more efficient and expedited manner.

Fig. 17 shows the results of diagnosing the operational signatures over an entire month based on data gathered at three-day intervals. Each bar illustrates the distribution ratio of a specific type throughout the month. When identifying the entire month's data using 1–3 days of accumulated data compared to using the full month's data, there was a detection of 1.1 % more instances of uncomfortable operation. Additionally, proper operation increased by 9.7 %, while energy waste types decreased by 8.1 %. Ultimately, 3.6 % of the total data was identified as having an uncomfortable indoor environment, comprising 1.4 % inefficient operation and 2.2 % uncomfortable operation. Additionally, recommendation for reduced operation was detected in 0.2 % of the

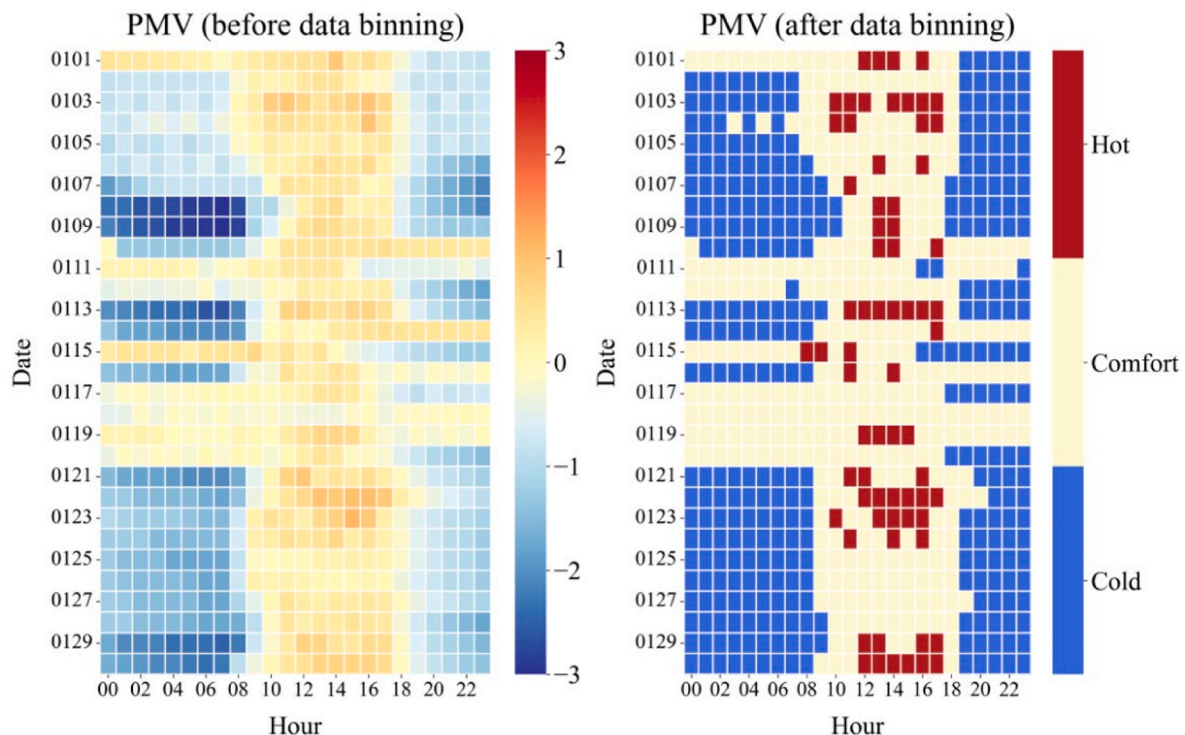


Fig. 15. Data binning using absolute criteria (PMV).

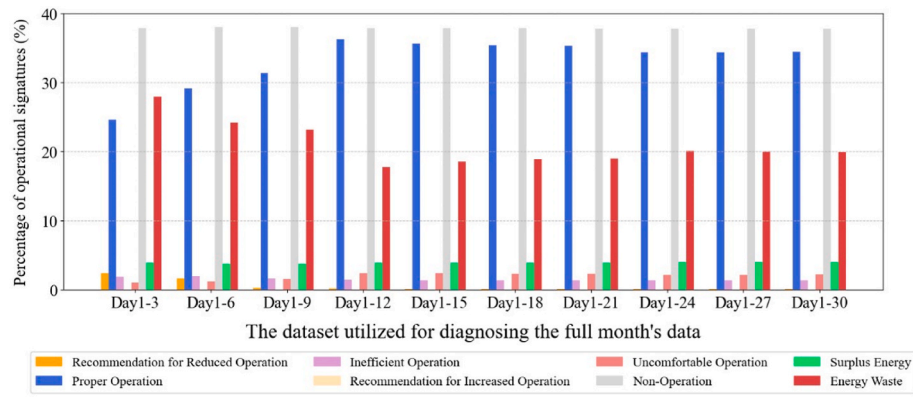


Fig. 17. Identification results of operational signature types for the full month's data using the HOS classifier based on 3-day cumulative data.

data. Overall, a high proportion of proper operation was observed, which is assumed to be because the target building is not large-scale and is well-managed. The most significant issue in the target space was the high rate of energy waste, which accounted for 19.9 %, primarily occurring during nighttime. This was primarily due to the continuous operation of the heating systems on specific days of the month, such as January 10th, 11th, and 12th, even during times deemed unoccupied (Fig. 18). In such cases, where energy waste patterns appear dominantly, integrating occupancy detection sensors with HVAC control algorithms may be an effective approach. However, the significance of the DT-BEMS proposed in this study lies in generating energy efficiency information by identifying diverse operational signatures throughout the building operation phase.

The results of such operational signature identifications contribute to generating real-time energy efficiency insights. For example, in cases of energy waste, DT-BEMS can provide information such as recommendations for turning off the building facility system to minimize further energy waste because the system is in operation while no occupants are present. Additionally, identifying the surplus energy type enables building users to explore ways to utilize the excess energy effectively. One of the approaches is to integrate with Smart Grid technology [36] to

facilitate energy trading between users or to implement energy management strategies through demand response (DR) [37]. By connecting this energy efficiency information to the building system, it is possible to reduce unnecessary energy consumption, improve indoor thermal comfort, and adjust equipment operation or set points accordingly. This allows for intuitive, building-level energy optimization, enhancing overall energy efficiency.

4.2.2. Comparison before and after virtual building model application

To examine the changes in operational signature identification results following the application of behavioral models, Fig. 19 compares the identification results before and after applying the PMV virtual model and the occupancy model using the entire month's data. In the case of Fig. 19(a), PMV values were calculated by PMV equation based on temperature and humidity data measured by the BEMS, without applying a PMV virtual model. Additionally, occupant presence was determined using only a typical occupancy schedule, without incorporating an occupancy model. Under these conditions, the verification results were dominated by non-operation (37.8 %) and proper operation (36.1 %), while energy waste (15.4 %) and uncomfortable operation (4.1 %) were also detected. When only PMV virtual sensor was applied,

Operational signatures diagnosis results of the full month

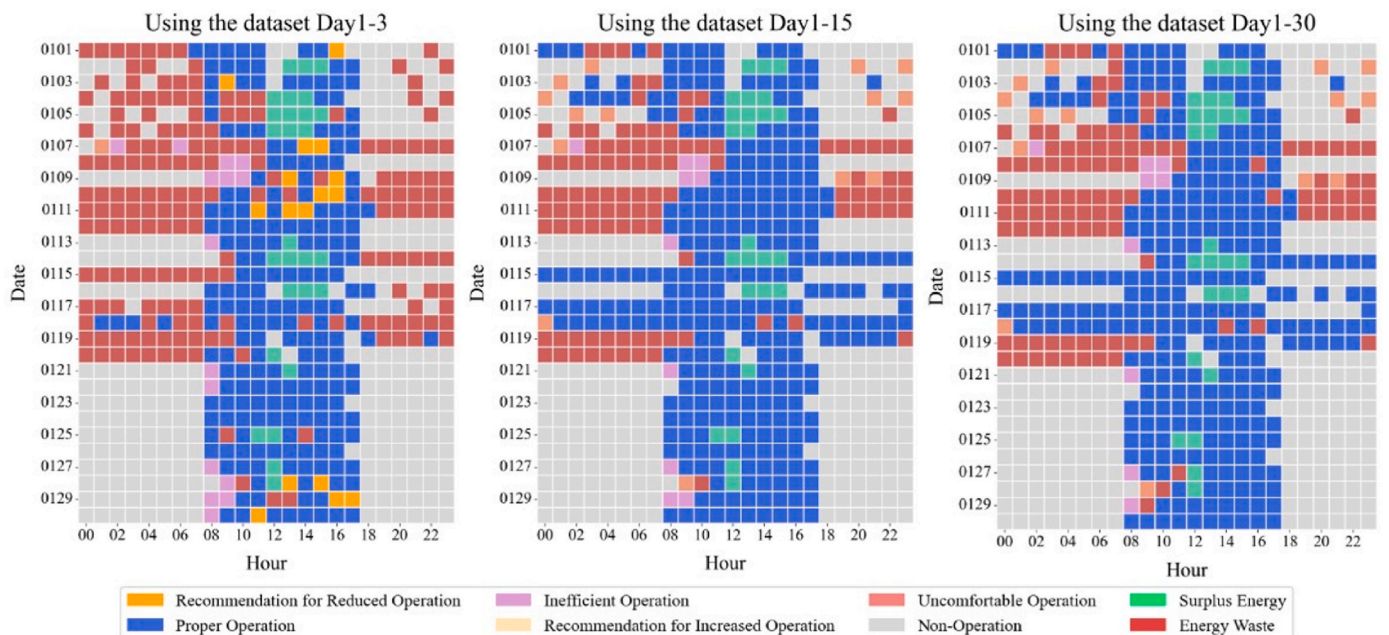


Fig. 18. Operational signature heatmaps.

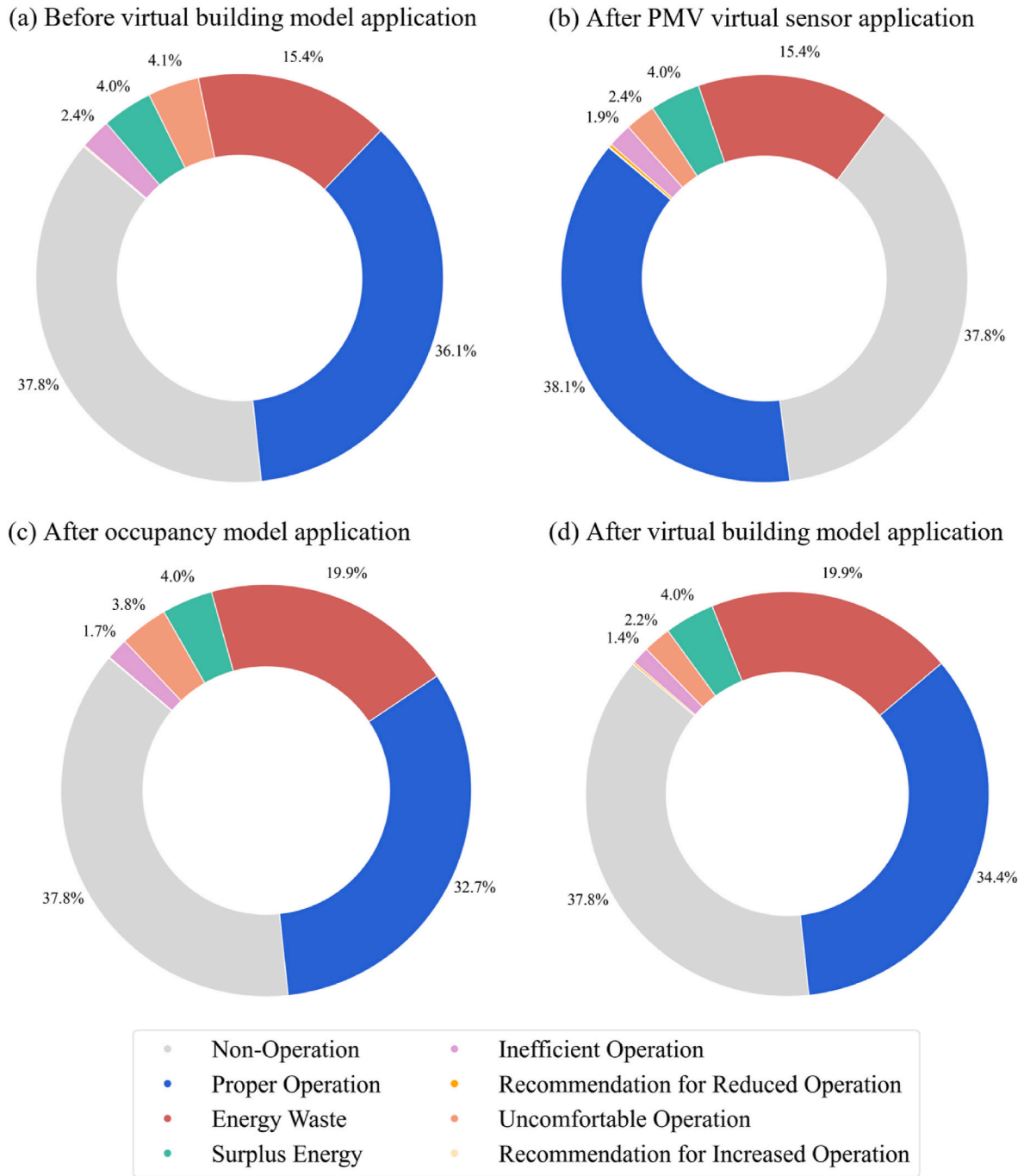


Fig. 19. Comparison of operational signatures distribution before and after virtual building model application.

compared to the pre-application results, uncomfortable operation decreased by 1.7 %, inefficient operation by 0.5 %, while proper operation increased by 2 %, and recommendations for reduced operation rose by 0.2 %. In the uncomfortable operation type, PMV values were 'cold,' while in proper operation, PMV values were 'comfort' or 'hot.' For the recommendation for reduced operation category, PMV values were 'hot.' This suggests that the PMV values for certain time periods noticeably increased with the application of the PMV virtual sensor. These results align with the trend identified in Fig. 13, where the PMV values increased after applying the virtual sensor. Since the PMV virtual sensor was modeled based on intrusive data, it accurately reflects the characteristics of the indoor environment. Therefore, it is evident that the changes in results after applying the PMV virtual model offer a more accurate identification by considering the thermal comfort of the real

occupied area. As demonstrated, the application of the PMV virtual model allows for a more precise real-time identification of operational signatures compared to when it was not applied. Moreover, a comparison was made between determining occupancy based only on a typical occupancy schedule and using the occupancy model. Proper operation decreased by 3.4 %, while inefficient operation and uncomfortable operation decreased by 0.7 % and 0.3 %, respectively. In contrast, the detection of energy waste increased by 4.5 %.

4.3. Essential requirements for the development of DT-BEMS

The implementation of DT-BEMS in the target space, based on the findings of this study, requires several essential conditions. First, at least 12 days of 24-h energy consumption, power generation, and indoor

temperature and humidity data must be available at a resolution higher than 1 h. As shown in Fig. 16, achieving a certain level of accuracy in the operational signature identification necessitates a minimum of 12 days of data. When data was accumulated for more than 12 days, the R-squared accuracy of the energy imputation models exceeded 0.8, demonstrating high reliability. The occupancy model was designed to require CO₂ concentration data for both daytime and nighttime. Therefore, the availability of 24-h data is necessary for its development. While the HOS classifier can be modeled regardless of data resolution, a resolution higher than 1 h is essential for heatmap-based visualization of the target space's operational signatures. Additionally, if the data resolution is lower than 1 h, the real-time aspect may be diminished, and 12 days of data may be insufficient for DT-BEMS. In the case of the PMV virtual model applied in this study, intrusive measurement data were utilized, making the availability of such data a prerequisite.

By utilizing these datasets, DT-BEMS energy efficiency strategies can be developed in Dynamo environment.

4.4. Limitations and future directions

This study has several limitations. The occupancy model applied to the target space determined occupancy status based on the operational schedule and CO₂ concentration in a constrained sensing environment. However, further refinement of the model is necessary. Additionally, the HOS classifier was designed with a focus on the characteristics of the target building, which employs an EHP-based heating and cooling system. Moreover, it was only applied for a month (January) during winter. Finally, while DT-BEMS effectively identified operational signatures and generated energy efficiency information, the study lacks a quantitative assessment of its impact on actual energy savings and operational cost reductions.

Considering these limitations, future research should proceed as follows. First, ground truth data of occupancy status should be established by integrating passive infrared (PIR) sensors or installing cameras in the target space, enabling the development of a more advanced occupancy model. Second, the HOS classifier that can be generally applied across various building systems should be developed and validated under diverse climate conditions to ensure adaptability. Third, DT-BEMS should be directly integrated with the building control system, incorporating an ontology-based AI agent system. By leveraging BEMS informatics, the AI agent can be specialized for the target building system and domain, thereby enabling building control based on operational signature types. This integration will facilitate the quantitative evaluation of energy savings and operational cost reductions. Additionally, integrating with control algorithms such as model predictive control (MPC) can further maximize energy-efficient operation.

5. Conclusion

In this study, the concept of DT-BEMS was proposed and implemented in the Dynamo environment to verify the operational signatures of the target space in real-time, thereby generating building energy efficiency information. In BEMS informatics, the representation of the BEMS data collection system and building structure in an ontology format was completed to promote the digitalization of both the building and the BEMS system. From the Digital Twin perspective, three types of virtual models were developed. The energy imputation model improved data quality by filling in missing values with data that reflect the building's physical behavior. The PMV virtual model enabled the estimation of PMV values within actual occupied zones. Additionally, an occupancy model was implemented to determine occupancy status based on operational schedules and CO₂ data. By integrating the outcomes of the virtual models with data binning for pattern analysis, the HOS classifier diagnosed the operational signatures of specific time periods. Through this approach, DT-BEMS generated information and knowledge about building energy efficiency, with updates occurring

every three days to improve accuracy and reliability. In the target building, using 12 days of data achieved 97.9 % similarity in identifying operational signatures compared to the results obtained using a full month's dataset. When analyzing the operational signatures over a full month, 23.5 % of operations were identified as inappropriate. Furthermore, by comparing operational signature diagnostics before and after applying the virtual models, the importance of the virtual model was emphasized.

In conclusion, this study successfully established an information system for analyzing building operational states by leveraging real-time multi-source data from BEMS through DT-BEMS. Generally, diagnosing a building's operational state and identifying appropriate response strategies require continuous expert-level decision-making in the building environment domain. The energy efficiency strategy proposed in this study is initially designed by experts and evolves autonomously during the operational phase. This approach has the potential to significantly reduce the cost and time required for building energy efficiency improvements. Furthermore, this system establishes a foundation for effective building control that considers indoor environmental conditions.

CRediT authorship contribution statement

Jaemin Hwang: Writing – original draft, Methodology, Investigation. **Jiwon Kim:** Software, Resources, Methodology. **Sungmin Yoon:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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